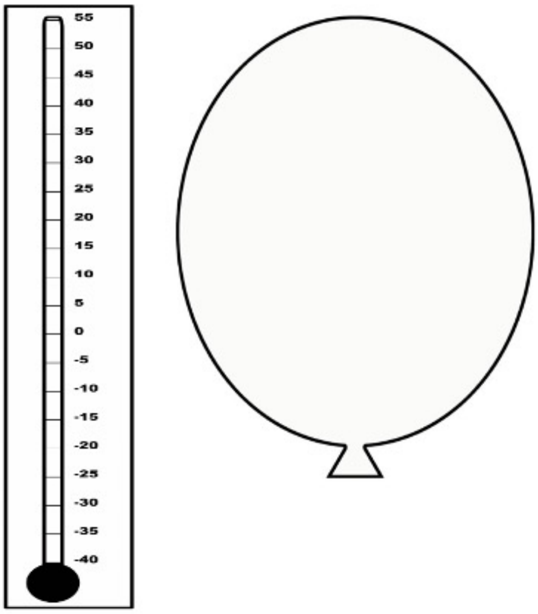
Name: ___________________________ Class:__________

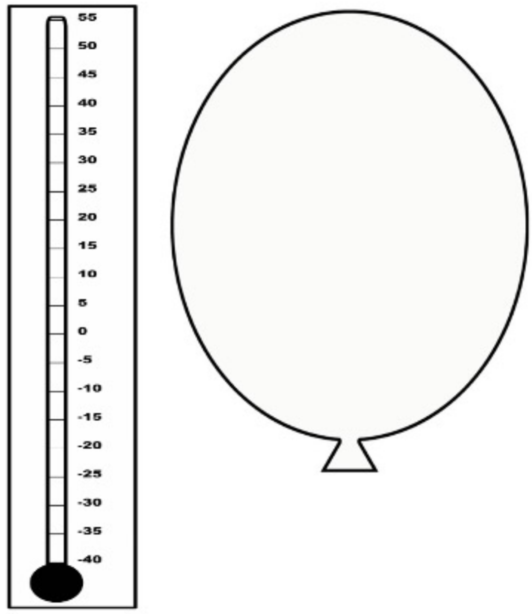
Properties of Matter End-of-Unit Study Guide

1. Draw on the thermometer to show the temperature and what the particles of gas are doing at that temperature.

5°C

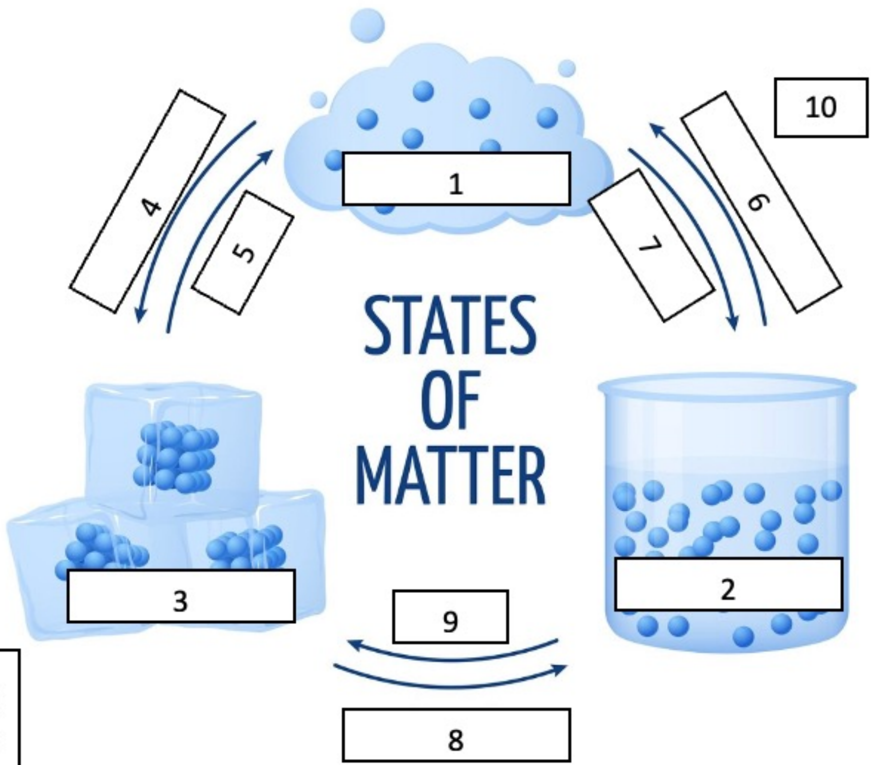


40°C



2. Use the word bank to complete the phase change diagram below:

1.__________________
2. _________________
3. _________________
4. _________________
5. _________________
6. _________________
7. _________________
8. _________________
9. _________________
10._______________



Word Bank

boiling point	freezing point	gas
sublimation	condensation	
melting point.	evaporation	
deposition	liquid	solid

3. Match the units, tools, and properties using the word bank below:

Word Bank

(mL) (m) (kg) (g) (L) (cm^3) ~~(g/mL)~~ (mm) ~~(g/cm^3)~~ scale ~~meter stick~~ measuring cup graduated cylinder ~~ruler~~ balance ~~kilograms~~ meters cubic centimeters ~~grams~~ liters millimeters ~~volume~~ mass weight density distance length

Unit Name	Unit Abbreviation	Measuring Tool	Property
grams, kilograms			
		meter stick, ruler	
			volume
	(g/mL) (g/cm^3)		

4. Use the word bank to complete the metric prefix table below:

Word Bank

milli-	kilo-	centi-	÷100	x1000	÷1000

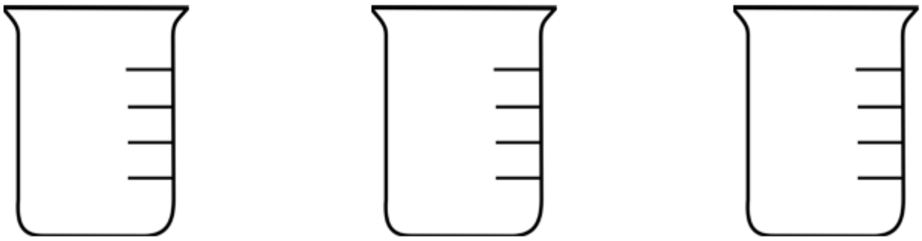
	x100	x10	x0	÷10		

5. Write the equations for finding density, mass, and volume. Use the equations to find the missing values.

Density = -----------   mass = ______ x ______   volume = -----------

mass	volume	density
100g	10mL	
1000g		1g/mL
	100L	40kg/L
500g		50g/cm^3
8g	1mL	

6. Draw a particle model for each phase of matter to show what the particles are doing at each phase.



solid   liquid   gas

For each statement, write **true or false**. **Correct or explain** each false statement on the line.

7. Chemical reactions produce new substances by increasing the size and number of atoms. ____

8. An element is made of molecules connected by chemical bonds and may have different properties than its parts. ______________________________

9. A solution is a homogenous mixture where all substances are evenly distributed, such as salt dissolved in water. ______________________

10. A compound is a mixture of atoms that have gone through a physical change. ____

11. An object that has a high volume will always have a high density.

____

12. Food rotting is an example of physical change because new, smelly substances are produced by micro-organisms and exposure to air.

____

13. Gold is an example of an element because it is a pure substance made of only one kind of atom. ____

14. Iron must be denser than gold because each individual iron atom is heavier, or has more mass, than each gold atom. ______________________________

15. Compounds like carbon dioxide can be separated into its parts using physical or mechanical processes, such as filtration or condensation.

____

16. A solute is insoluble when it cannot be dissolved in a solvent. ______________

17. An object has buoyancy when its density is greater than the substance below it, causing it to float. ____

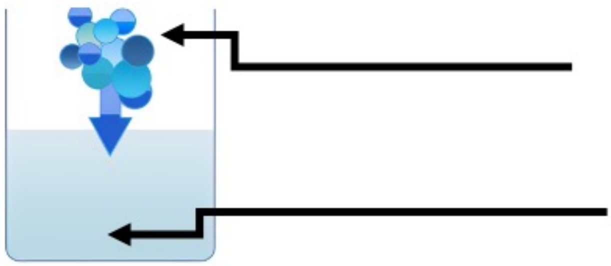
18. Liquid water, ice, and water vapor are examples of three different kinds of compounds. ____

19. A solution is a homogenous mixture. ___________________________________

20. Label the parts of a solution:

Word Bank

soluble
solute
solvent
insoluble



Directions: Use the period square to answer the questions.

[Ar]4s¹ 19

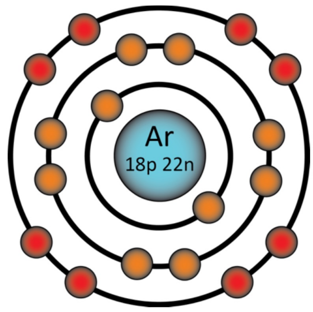
K

Potassium
39.0983

1. What is the atomic symbol for potassium? ________
2. How many protons does a potassium atom have? ______
3. How many neutrons does an average potassium atom have? ________
4, How many valence electrons does a potassium atom in its ground state have? ________

5. Draw a Bohr model of a potassium atom in the space provided. Remember to include
- The number of protons
- The number of neutrons
- Electron shells
- Valence electrons

Example:



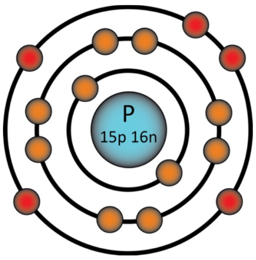
Directions: Use the word bank to complete the chart. Some words may be used twice.

<u>**Word Bank**</u> Halogens Alkaline Earth Metals Transition Metals Group 18 Group 1 form anions(-) are less reactive have complete electron shells form cations (+) Semiconductors Lanthanides and Actinides post-transition metals radioactive

Series	Location on Periodic Table	Properties
	Middle of the Periodic Table	
Metaloids	On the right side of the Periodic Table, Groups 13-17	
	Group 17	
Noble Gasses		
	Group 2	
	Between Transition metals and metaloids (semi-metals)	Soft, brittle, low melting points
	Usually found at the bottom of the Periodic Table	
Non-metals	Includes Carbon, Nitrogen, Oxygen, and the Halogens	
Alkalai Metals		

Directions: Use the Bohr model to answer the questions.

Phosphorus



1. How many **valence electrons** does Phosphorus have? ________
2. How many **neutrons** does Phosphorus have? ________
3. How many **protons** does Phosphorus have? ________
4. How many total **electrons** does phosphorus have? ________

Use the word bank to complete the diagram & chart.
Words may be used more than once. .

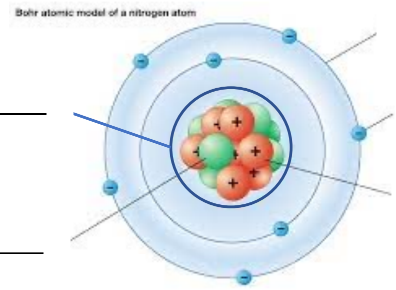
Word Bank

proton   electron orbital (shell)

electron   neutron   nucleus

____

____



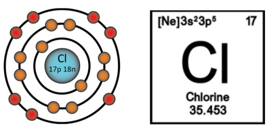
____

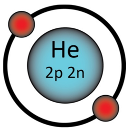
____

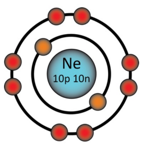
____

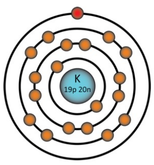
Structure	Description
	Energy level or "shell" where electrons may be found
	Positivly charged particle
	Center of the atom where protons and neutrons are held together by the Strong Nuclear Force
	Negatively charged particle
	Uncharged sub-atomic particle
	The smallest of the subatomic particles. Moves at nearly the speed of light
	Gaining or losing these will form a charged **ion**
	Gaining or losing these will form an **isotope**
	Gaining or losing these will form a new element

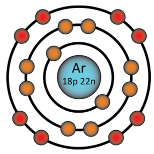
Which is most likely to react with Chlorine?



A. 

B. 

C. 

D. 

Explain your reason: ____

If an atom has 4 **protons**, 5 **neutrons**, and 4 **electrons**, what is its atomic mass? _____u

Show how each substance is classified. There may be 2 answers or none.

Ni CO_2 Cl_2 NaCl

$C_6H_{12}O_6$.   Hg O_3

Neon   Bronze   Water

Substance	Examples
Element	
Molecule	
Compound	

Practice using a periodic table to complete the sentences below:

1. The atomic symbol for lead is _____________.
2. The Halogen in Period 4 (or D) is _______________________.
3. The lightest element in Group 15 is _______________________.
4. C, P, Se, Cl, and Ne are all _________________________.
5. One example of an unstable, or radioactive element is _________________.
6. Four examples of metals are ____.
7. The atomic number of Gold is _________.
8. Two examples of Actinides are _____________________________________.
9. On the periodic table, groups go ____.
10. On the periodic table, periods go ____.

Use your notes and precious assignments to review the following:

11. Describe the difference between a physical change and a chemical change with one example of each,

____

____

____

____

____

12. Describe the particles of each state of matter.

Solid: ____

____

Liquid: ____

____

Gas: ____

____

Remember to review all vocabulary words and review notes in your science notebook, making sure everything is easy to find and read!